

TASK 2: EXPLORATORY DATA ANALYSIS (EDA)

INTRODUCTION

Exploratory Data Analysis (EDA) was performed on the sales dataset to understand the data structure, identify missing values, analyze statistical patterns, and discover relationships between different variables.

The main objective of this project is to clean the dataset, analyze important features, and visualize the data distribution to extract meaningful insights.

Tools:

- Python
 - Visual Studio Code
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Excel Dataset
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DATASET DESCRIPTION

The dataset contains sales-related information used for analyzing product performance, category-wise sales, profit distribution, and regional patterns.

The dataset includes attributes such as:

- Product Name
- Category
- Sales
- Profit
- Region

The dataset was used to perform data analysis and identify important patterns.

STEPS PERFORMED

1. Imported required libraries

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were imported for data analysis and visualization.

2. Loaded dataset

The sales dataset was imported into Python using Pandas for further processing and analysis.

3. Explored dataset structure

The dataset structure was analyzed by checking:

- Number of rows and columns
 - Data types
 - Dataset information
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4. Checked missing values

The dataset was examined for missing or empty values to improve data quality.

5. Cleaned the dataset

Duplicate and unnecessary data values were identified and handled to prepare the dataset for analysis.

6. Performed statistical analysis

Statistical calculations were performed to understand:

- Average values
 - Minimum values
 - Maximum values
 - Data distribution
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7. Created visualization plots

Different visualizations were created using Matplotlib and Seaborn, including:

- Sales Distribution Plot
 - Category Analysis Chart
 - Profit Analysis Graph
 - Correlation Heatmap
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8. Identified patterns and insights

The analysis helped identify sales trends, category performance, and relationships between numerical variables.

OBSERVATIONS

- The dataset contains multiple sales-related attributes.
 - Data cleaning improved dataset quality.
 - Statistical analysis provided information about sales and profit patterns.
 - Visualization helped understand data distribution and comparisons.
 - Different categories showed variations in sales performance.
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CONCLUSION

Exploratory Data Analysis was successfully performed on the sales dataset using Python.

The analysis helped in understanding the dataset structure, handling data quality issues, identifying important patterns, and visualizing relationships between different attributes.

The insights obtained from EDA can be further used for data visualization, decision-making, and advanced analytics tasks.

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